

ASS 3 - Subnetting

```
#include <stdio.h>
```

```
int main() {
```

```
    int a, b, c, d;
```

```
    printf("Enter IP Address: "); //192.168.10.5
```

```
    scanf("%d.%d.%d.%d", &a, &b, &c, &d);
```

```
    if (a < 0 || a > 255 ||
```

```
        b < 0 || b > 255 ||
```

```
        c < 0 || c > 255 ||
```

```
        d < 0 || d > 255) {
```

```
        printf("\nInvalid IP Address!\n");
```

```
        return 0;
```

```
    }
```

```
    if (a >= 0 && a <= 127) {
```

```
        printf("\nClass : A\n");
```

```
        printf("IP Range : 0.0.0.0 - 127.255.255.255\n\n");
```

```
        printf("Subnet Mask : 255.0.0.0\n");
```

printf("A subnet mask is a 32-bit number used with an IP address to identify which part of the address represents the network and which part represents the host (device).\n\n");

```
        printf("\n32-bit Representation\n");
```

```
        printf("Network ID\n");
```

```
        printf("[0][N][N][N][N][N][N][N]");
```

```
        printf("[H][H][H][H][H][H][H][H]");
```

```
        printf("[H][H][H][H][H][H][H][H]");
```

```
        printf("[H][H][H][H][H][H][H][H]\n\n");
```

```
        printf("Hosts per Network ID :  $2^{24} - 2 = 16,777,214$ \n");
```

```
        printf("Calculation :\n");
```

```
        printf("Host bits = 24\n");
```

```
        printf(" $2^{24} = 16,777,216$ \n");
```

```
        printf("Subtract 2 (Network ID and Broadcast Address)\n");
```

```
        printf("= 16,777,214 Hosts\n\n");
```

```
        printf("Number of Networks :  $2^7 = 128$ \n");
```

```
        printf("Calculation :\n");
```

```
        printf("Network bits = 7\n");
```

```
        printf(" $2^7 = 128$  Networks\n\n");
```

```

if (a == 127)
    printf("Loopback Address : YES\n");
else
    printf("Loopback Address : NO\n");
    printf("A loopback address is a special IP address that a computer uses to
communicate with itself for testing and troubleshooting network software
without sending data over a physical network.\n");
    printf("The purpose of a loopback address is to allow a computer to
communicate with itself without using a physical network connection.\n");
} else if (a >= 128 && a <= 191) {
    printf("\nClass : B\n");
    printf("IP Range : 128.0.0.0 - 191.255.255.255\n\n");

    printf("Default Subnet Mask : 255.255.0.0\n");
    printf("A subnet mask is a 32-bit number used with an IP address to
identify which part of the address represents the network and which part
represents the host (device).\n\n");
    printf("\n32-bit Representation\n");
    printf("[1][0][N][N][N][N][N][N]");
    printf("[N][N][N][N][N][N][N][N]");
    printf("[H][H][H][H][H][H][H][H]");
    printf("[H][H][H][H][H][H][H][H]\n\n");

    printf("Hosts per Network ID : 2^16 - 2 = 65,534\n");
    printf("Calculation : \n");
    printf("Host bits = 16\n");
    printf("2^16 = 65,536\n");
    printf("Subtract 2 (Network ID and Broadcast Address)\n");
    printf("= 65,534 Hosts\n\n");

    printf("Number of Networks : 2^14 = 16,384\n");
    printf("Calculation : \n");
    printf("Network bits = 14\n");
    printf("2^14 = 16,384 Networks\n\n");

    printf("Loopback Address : NO\n");
    printf("A loopback address is a special IP address that a computer uses to
communicate with itself for testing and troubleshooting network software
without sending data over a physical network.\n");
    printf("The purpose of a loopback address is to allow a computer to
communicate with itself without using a physical network connection.\n");
} else if (a >= 192 && a <= 223) {
    printf("\nClass : C\n");
    printf("IP Range : 192.0.0.0 - 223.255.255.255\n\n");

    printf("Default Subnet Mask : 255.255.255.0\n");

```

```
printf("A subnet mask is a 32-bit number used with an IP address to  
identify which part of the address represents the network and which part  
represents the host (device).\n\n");
```

```
printf("\n32-bit Representation\n");  
printf("[1][1][0][N][N][N][N][N]");  
printf("[N][N][N][N][N][N][N][N]");  
printf("[N][N][N][N][N][N][N][N]");  
printf("[H][H][H][H][H][H][H][H]\n\n");
```

```
printf("Hosts per Network ID :  $2^8 - 2 = 254$ \n");  
printf("Calculation :\n");  
printf("Host bits = 8\n");  
printf(" $2^8 = 256$ \n");  
printf("Subtract 2 (Network ID and Broadcast Address)\n");  
printf("= 254 Hosts\n\n");
```

```
printf("Number of Networks :  $2^{21} = 2,097,152$ \n");  
printf("Calculation :\n");  
printf("Network bits = 21\n");  
printf(" $2^{21} = 2,097,152$  Networks\n\n");
```

```
printf("Loopback Address : NO\n");  
printf("A loopback address is a special IP address that a computer uses to  
communicate with itself for testing and troubleshooting network software  
without sending data over a physical network.\n");
```

```
printf("The purpose of a loopback address is to allow a computer to  
communicate with itself without using a physical network connection.\n");
```

```
} else if (a >= 224 && a <= 239) {  
printf("\nClass : D\n");  
printf("IP Range : 224.0.0.0 - 239.255.255.255\n\n");
```

```
printf("Purpose : Multicast\n\n");
```

```
printf("Default Subnet Mask : Not Applicable\n");  
printf("A subnet mask is a 32-bit number used with an IP address to  
identify which part of the address represents the network and which part  
represents the host (device).\n\n");
```

```
printf("Hosts per Network ID : Not Applicable\n");  
printf("Number of Networks : Not Applicable\n\n");
```

```
printf("Loopback Address : NO\n");  
printf("A loopback address is a special IP address that a computer uses to  
communicate with itself for testing and troubleshooting network software  
without sending data over a physical network.\n");  
printf("The purpose of a loopback address is to allow a computer to
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communicate with itself without using a physical network connection.\n");
    } else {
        printf("\nClass : E\n");
        printf("IP Range : 240.0.0.0 - 255.255.255.255\n\n");

        printf("Purpose : Experimental\n\n");

        printf("Default Subnet Mask : Not Applicable\n");
        printf("A subnet mask is a 32-bit number used with an IP address to
identify which part of the address represents the network and which part
represents the host (device).\n\n");

        printf("Hosts per Network ID : Not Applicable\n");
        printf("Number of Networks : Not Applicable\n\n");

        printf("Loopback Address : NO\n");
        printf("A loopback address is a special IP address that a computer uses to
communicate with itself for testing and troubleshooting network software
without sending data over a physical network.\n");
        printf("The purpose of a loopback address is to allow a computer to
communicate with itself without using a physical network connection.\n");
    }

    return 0;
}

```